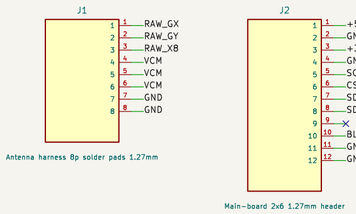
PF233-AB1A: AB1A receive analog board: 3 channels (GX, GY, X8), blanking, gain, 16-bit quad ADC (AD7389-4)

How to read: parts sit in functional blocks (dashed frames). Every pin ends in a short wire with its NET NAME: equal names are connected (also between blocks). A cross = intentionally unconnected pin.
Diode symbols point anode to cathode. Pin names are those of the library; items marked VERIFY in docs/ASSUMPTIONS.md are unverified. MODELED design: not built, not bench-tested.

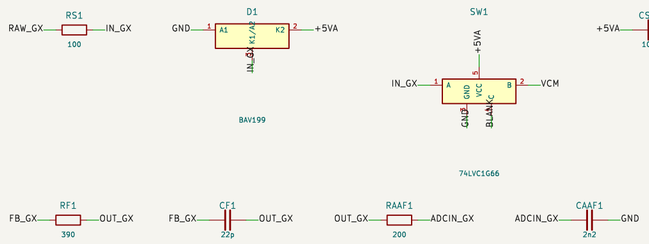
1 Connectors

J1: antenna harness (raw coil signals RAW_*, and VCM returns).
J2: main-board 2x6 header: +5VA, +3V3D, SPI (SCLK, CS, SDO*), BLANK.



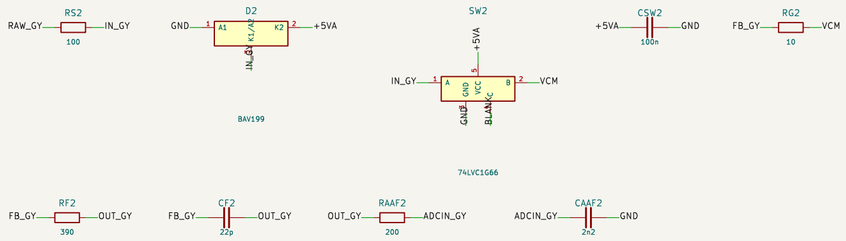
2 Channel GX

RAW_GX -> RS1 (series) -> D1 (BAV199 clamp to +5VA / GND) -> SW1 (74LVC1G66; BLANK opens the path) -> IN_GX -> op-amp U1 half A.
Non-inverting gain stage: RG1 to VCM, RF1 || CF1 feedback; OUT_GX -> RAAF1 / CAAF1 anti-alias -> ADCIN_GX -> UA1 input pair 1 (AD7389-4, differential: AIN+ = ADCIN_GX, AIN- = VCM).
The shared AD7389-4 is in block 7.



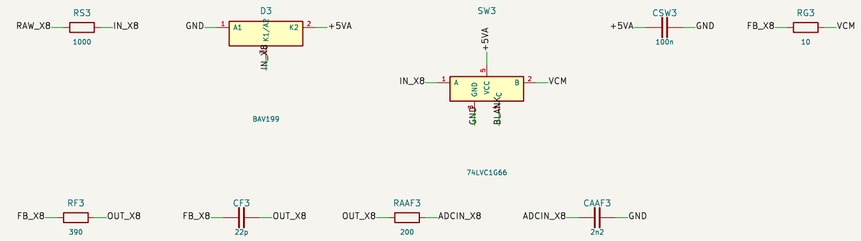
3 Channel GY

RAW_GY -> RS2 (series) -> D2 (BAV199 clamp to +5VA / GND) -> SW2 (74LVC1G66; BLANK opens the path) -> IN_GY -> op-amp U1 half B.
Non-inverting gain stage: RG2 to VCM, RF2 || CF2 feedback; OUT_GY -> RAAF2 / CAAF2 anti-alias -> ADCIN_GY -> UA1 input pair 2 (AD7389-4, differential: AIN+ = ADCIN_GY, AIN- = VCM).
The shared AD7389-4 is in block 7.



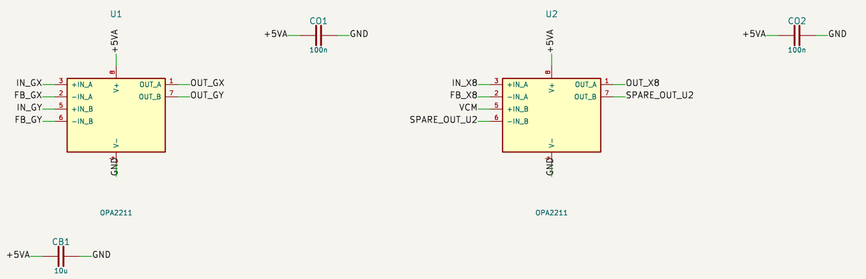
4 Channel X8

RAW_X8 -> RS3 (series) -> D3 (BAV199 clamp to +5VA / GND) -> SW3 (74LVC1G66; BLANK opens the path) -> IN_X8 -> op-amp U2 half A.
Non-inverting gain stage: RG3 to VCM, RF3 || CF3 feedback; OUT_X8 -> RAAF3 / CAAF3 anti-alias -> ADCIN_X8 -> UA1 input pair 3 (AD7389-4, differential: AIN+ = ADCIN_X8, AIN- = VCM).
The shared AD7389-4 is in block 7.



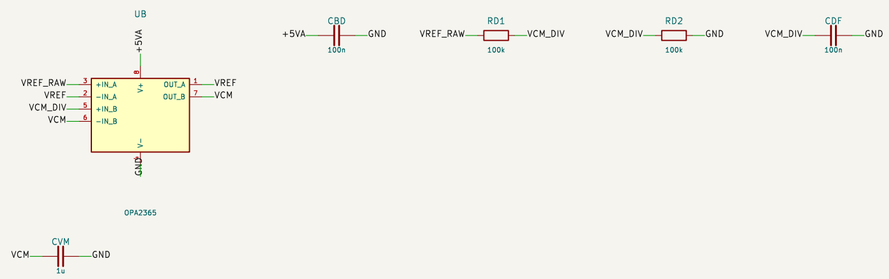
5 Gain-stage op-amps

OPA2211 dual op-amps (each half = one channel) from +5VA, decoupled CO*; CB1 bulk.



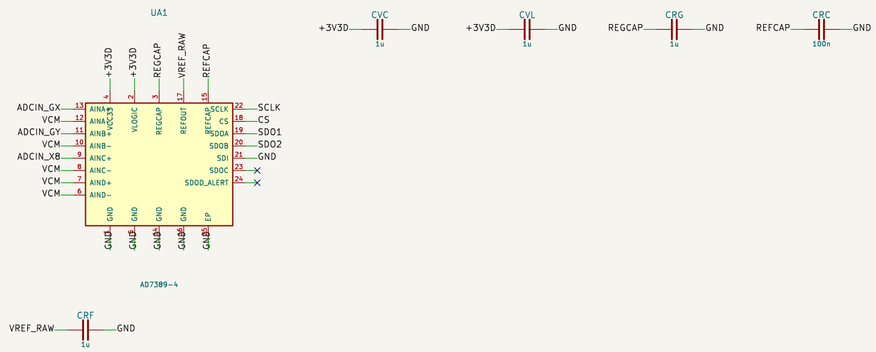
6 Reference and common-mode

UA1 REFOUT (internal 2.5 V) -> VREF_RAW (CRF 1 uF); RD1 / RD2 (100 k) divide it to VCM_DIV (1.25 V) -> UB (OPA2365) half B -> VCM (CVM); UB half A is parked as a follower.



7 Quad simultaneous ADC

UA1 AD7389-4 (16-bit, 2 MSPS, four differential inputs, internal reference): inputs ADCIN_* with VCM; unused pairs are parked on VCM.
VCC and VLOGIC from +3V3D (CVC, CVL 1 uF), REGCAP (CRG), REFCAP (CRC), REFOUT (CRF). 2-wire readout: SDOA (channels A then C) to J2 SDO1, SDOB to J2 SDO2; SDI tied to GND; CS starts a conversion.



Generated: src/pcb/sch_func.py; checked against the netlist.
MODELED design: not built, not bench-tested.
AB1A receive analog board: 3 channels (GX, GY, X8), blanking, gain, 16-bit quad ADC (AD7389-4)
ARX / PHI-FOLD 233

Sheet: /		
File: PF233-AB1A_functional.kicad_sch		
Title: PF233-AB1A functional schematic		
Size: User	Date: 2026-10-26	**Rev: r3**
KiCad E.D.A. 9.0.9		Id: 1/1